

The pair singular series of a polynomial V : the dilated pieces below $H^{1/2}$

Jasper Reichardt*

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Abstract

[Draft. The proofs of Theorems 3 and 4 are in `research/explore/PAPER-V-thmM.md` and `PAPER-V-smallend-draft.md` and are under adversarial reading; nothing here is to be circulated before those readings are recorded.] For a monic irreducible quadratic f without fixed prime divisor, part III decomposed the Cesàro off-diagonal $\text{Off}_f^*(H)$ into pieces indexed by the dilation u and proved that the pieces with $u > H^{1/2+\varepsilon}$ contribute a main term $c_{\text{off}}(f)H$. This paper treats dilations below that threshold. We prove that the pieces with $H^{0.47} < u \leq H^{0.53}$ contribute $O(H^{1-\delta})$, so that $\sum_{u > H^{0.47}} w(u) \mathcal{P}_u(H/u) = c_{\text{off}}(f)H + O(H^{1-\delta})$; the input is a bilinear estimate for Kloosterman fractions of Bourgain and Garaev together with a bound for the additive energy of reciprocals, valid for every modulus, which we deduce from an argument of Cilleruelo and Garaev. For negative discriminants we prove a second result at the other end: the pieces with $u \leq H^{1/3-c}$ contribute $O(H^{1-\delta(c)})$, by running the automorphic kernel method of Grimmelt and Merikoski with the leading coefficient u^2 tracked. What remains is a neighbourhood of $u = H^{1/3}$.

1 Introduction

[to write: the picture in exponent coordinates; what part III leaves open; the two ends; the band at $H^{1/3}$ and what it would take (PROOFS §31–§34).]

2 Notation

[import from part III §1–§2: w, λ, ω , admissible moduli, $\mathcal{P}_u, G_{d'}, \bar{G}_{d'}, G^\circ, \rho_k, \xi_k$.]

3 The energy of reciprocals and bilinear Kloosterman fractions

Lemma 1 (energy of reciprocals). [*PAPER-V-thmM.md Lemma E; proof written, two readings.*]

Lemma 2 (Bourgain–Garaev). [*cite [1]; Corollary B'.*]

4 The pieces around $u = H^{1/2}$

Theorem 3. [*Theorem M; proof = Lemmas K, E, B', S and Steps R1–R5.*]

*Independent researcher. Correspondence: jasperreichardt@gmail.com.

5 The small dilations

Theorem 4. *[Theorem S ($D < 0$); proof = S1–S8 with Lemma A'', S1', S2', O5, O6'.]*

6 What remains

[the band at $u = H^{1/3}$: the two mechanisms degenerate there; the accounting of PROOFS §31–§34; what a positive saving at the point would give (A_f), and what a point gap with polylog losses would give (part I's Hypothesis (E)).]

References

- [1] Jean Bourgain and Moubariz Z. Garaev. Kloosterman sums in residue rings. *Acta Arith.*, 164(1):43–64, 2014.